

Lesson 22 — Loops with Moves & Sounds (+ Fix-It)

Matches LEGO Teacher Portal lesson: "Play All Day".

Objective (I can...): I can develop a program that puts a few actions inside one loop so a whole pattern repeats. **CS concept:** Developing a program with loops (a loop body with more than one action) · **Builds on:** Lesson 21 (single-action repeat).

AI Connection (2 min, read aloud): Each training "round" repeats the same steps on new examples until the AI gets better. *Ask the group:* "What might get a little better each time the loop runs again?"

Materials: LEGO CS & AI kit + device per pair. **Model reuse:** Loops model from Lesson 21 — code only.

60-minute flow - Warm-up (10): Pattern clap — coach does "clap-stomp-clap-stomp"; kids find the chunk that repeats and do it as a group. That chunk goes in the loop. - **Concept + demo (8):** Build a loop with move + sound + wait; run it and point out the whole chunk repeats. - **Build/Code (25):** Pairs develop a 3-action repeating pattern inside one loop (spin → beep → wait). **Fix-It mini-task (3 min):** hand each pair a "silly" program where a block sits OUTSIDE the loop — they find it and drag it in so the whole pattern repeats. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Add a light or a backward move to the loop; run it again. - **Share + tidy-up (10):** Pairs show the pattern and name the actions inside the loop. Model stays built.

Pair roles: Driver codes the loop, Navigator names the actions to include. Swap at the wiggle break. *(Trio: third leads the Fix-It hunt.)*

Differentiation: *Support* — a 2-action loop with a twin example. *Extension / fast-finisher* — a 4-action loop. *Reteach* — build the actions outside a loop first, confirm they work, then drag them inside together.

Exit check: Pair runs a loop whose body has 3+ actions that repeat together. **At-home (optional):** Make up a 3-move dance; repeat it 4 times for a grown-up. ~5 min.